Inference often combines conditioning and summarizing

Observed		**Condition**		**Summarize**
y	$\rightarrow$	$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z\mid y}, \Sigma_{z\mid y}\right)$	$\rightarrow$	$\hat{z} = E[z \mid y] = \mu_{z\mid y}$

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y)\, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z\mid y}$